Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Internal energy of n_1 mol of hydrogen of temperature T is equal to the internal energy of n_2 mol of helium at temperature $2T$. The ratio n_1/n_2 is

(A) $\frac{3}{5}$

(B) $\frac{2}{3}$

(C) $\frac{6}{5}$

(D) $\frac{3}{7}$

Q.2

The specific heats of argon at constant pressure and constant volume are $525\ \mathrm{J/kg}$ and $315\ \mathrm{J/kg}$, respectively. Its density at NTP will be

(A) $1.77\ \mathrm{kg/m^3}$

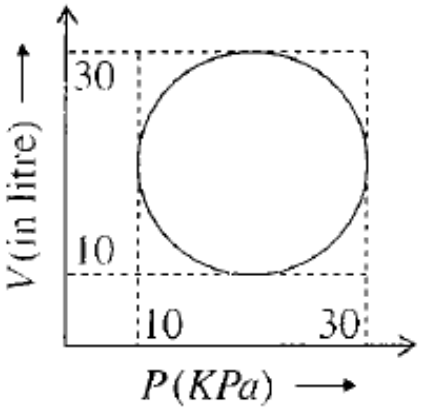
(B) $0.77\ \mathrm{kg/m^3}$

(C) $1.77\ \mathrm{g/m^3}$

(D) $0.77\ \mathrm{g/m^3}$

Q.3

Heat energy absorbed by a system in going through a cyclic process shown in Fig. is



(A) $10^7\pi\mathrm{J}$

(B) $10^4\pi\mathrm{J}$

(C) $10^8\pi\mathrm{J}$

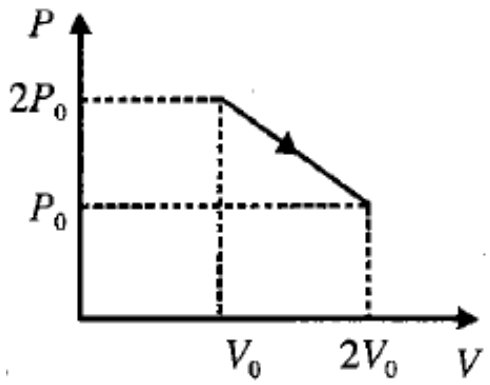
(D) $10^{-3}\pi\mathrm{J}$

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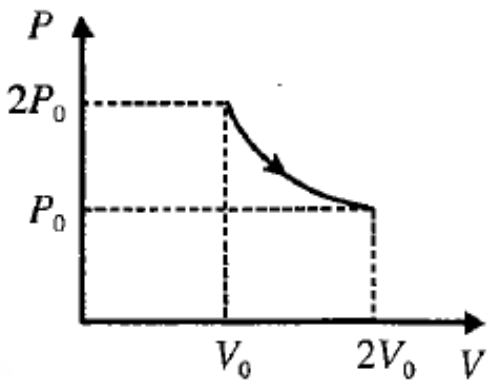
Q.4

The $P-V$ equation for a process of an ideal gas is given as
$P=\frac{12P_0}{V_0}V-\frac{4P_0}{V_0^2}V^2-7P_0$
The graphical representation of the above process is shown as

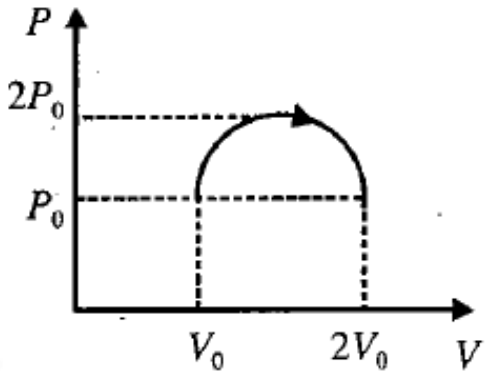
(A)



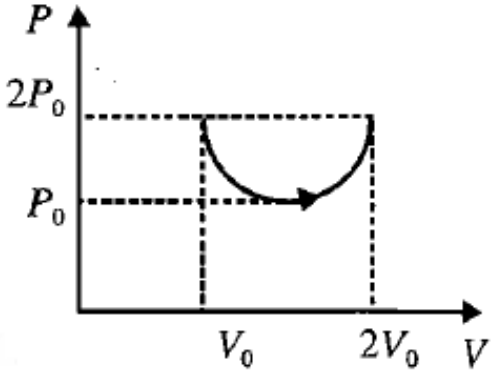
(B)



(C)



(D)



Q.5

A spherical balloon contains air at temperature T_0 and pressure P_0. The balloon material is such that the

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instantaneous pressure inside is proportional to the square of the diameter. When the volume of the balloon doubles as a result of heat transfer, the expansion follows the law

(A) $PV = \text{constant}$

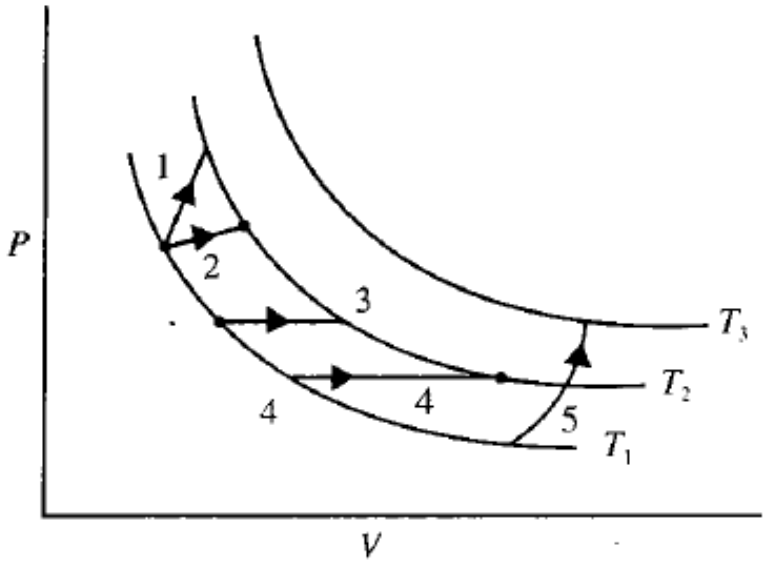
(B) $PV^{2/5} = \text{constant}$

(C) $PV^{-1} = \text{constant}$

(D) $PV^{-2/3} = \text{constant}$

Q.6

Figure 2.131 shows five paths traversed by a gas on a $P - V$ diagram. $\Delta U_1, \Delta U_2, \Delta U_3, \Delta U_4$ and ΔU_5 are the change in internal energy of the gas in paths $1, 2, 3, 4$ and 5, respectively. Then



(A) $\Delta U_5 > \Delta U_3$

(B) $\Delta U_3 > \Delta U_5$

(C) $\Delta U_1 > \Delta U_2$

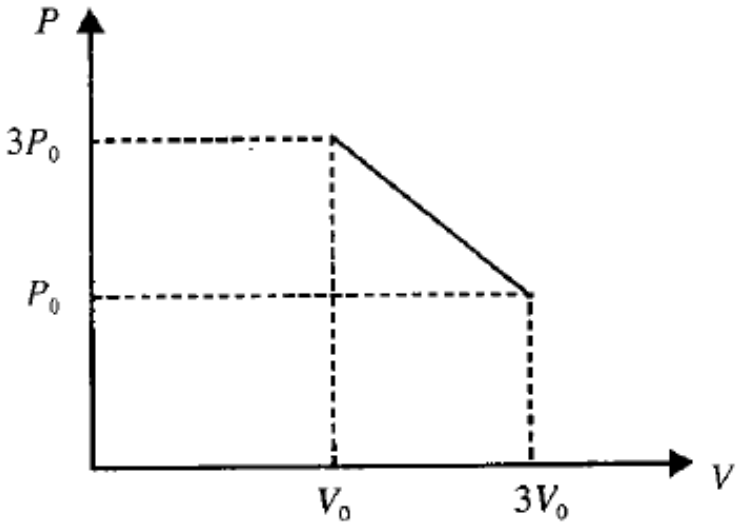
(D) $\Delta U_2 > \Delta U_5$

Q.7

n moles of gas in a cylinder under a piston is transferred infinitely slowly from a state with a volume of V_0 and a pressure $3P_0$ to a state with a volume of $3V_0$ and a pressure P_0 as shown in Fig. The maximum temperature

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that the gas will reach in this process is



(A) $\frac{P_0V_0}{nR}$

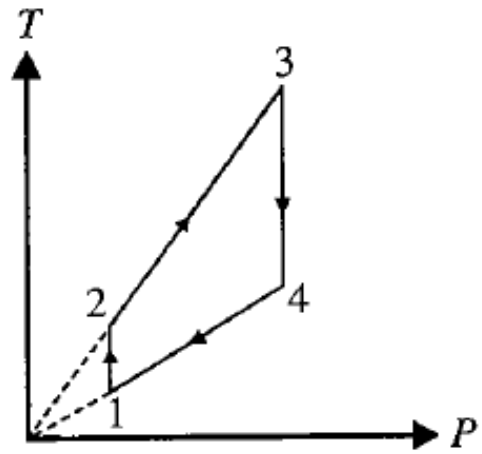
(B) $\frac{3P_0V_0}{nR}$

(C) $\frac{4P_0V_0}{nR}$

(D) $\frac{2P_0V_0}{nR}$

Q.8

Two moles of an ideal monatomic gas undergoes a cyclic process as shown in Fig. The temperatures in different states are given as $6T_1 = 3T_2 = 2T_4 = T_3 = 1800\ \text{K}$. Determine the work done by the gas during the cycle.



(A) $-10\ \text{kJ}$

(B) $-20\ \text{kJ}$

(C) $-15\ \text{kJ}$

(D) $-30\ \text{kJ}$

Q.9

Three samples of the same gas A, B and $C(\gamma = 3/2)$ initially have equal volume. Now the volume of each sample is doubled. The process is adiabatic for A, isobaric for B and isothermal for C. If the final pressures are equal for all the three samples, the ratio of their initial pressures is

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(A) $2\sqrt{2} : 2 : 1$

(B) $2\sqrt{2} : 1 : 2$

(C) $\sqrt{2} : 1 : 2$

(D) $2 : 1 : \sqrt{2}$

Q.10

If for hydrogen $C_p - C_v = m$ and for nitrogen $C_p - C_v = n$, where C_p and C_v refer to specific heats per unit mass respectively at constant pressure and constant volume, the relation between m and n is (molecular weight of hydrogen $= 2$ and molecular weight of nitrogen $= 14$)

(A) $n = 14m$

(B) $n = 7m$

(C) $m = 7n$

(D) $m = 14n$

Q.11

In an adiabatic process pressure is increased by $2/3\%$ if $\mathrm{C}_p/\mathrm{C}_V = 3/2$. Then the volume decreases by about

(A) $\frac{4}{9}\%$

(B) $\frac{2}{3}\%$

(C) 4%

(D) $\frac{9}{4}\%$

Q.12

A container of volume $1\ \mathrm{m}^3$ is divided into two equal compartments by a partition. One of these compartments contains an ideal gas at $300\ \mathrm{K}$. The other compartment is vacuum. The whole system is thermally isolated from its surroundings. The partition is removed and the gas expands to occupy the whole volume of the container. Its temperature now would be

(A) $300\ \mathrm{K}$

(B) $250\ \mathrm{K}$

(C) $200\ \mathrm{K}$

(D) $10\ \mathrm{K}$

Q.13

An ideal heat engine has an efficiency η. The coefficient of performance of the engine when driven backward will be

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(A) $1-(1/\eta)$

(B) $\eta/(1-\eta)$

(C) $(1/\eta)-1$

(D) $1/(1-\eta)$

Q.14

A gas is at 1 atm pressure with a volume $800\ \mathrm{cm}^3$. When $100\ \mathrm{J}$ of heat is supplied to the gas, it expands to $1\ \mathrm{L}$ at constant pressure. The change in its internal energy is

(A) $80\ \mathrm{J}$

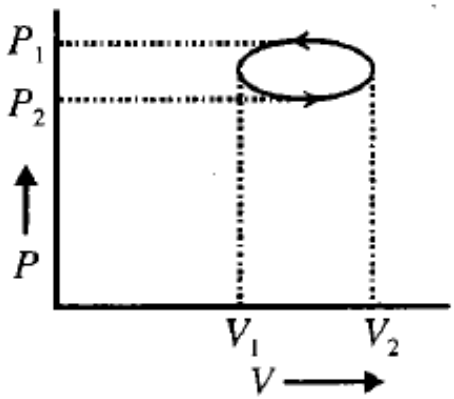
(B) $-80\ \mathrm{J}$

(C) $20\ \mathrm{J}$

(D) $-20\ \mathrm{J}$

Q.15

In the given elliptical $P-V$ diagram



(A) the work done is positive

(B) the change in internal energy is non-zero

(C) the work done $=-(\pi/4)\left(P_2-P_1\right)\left(V_2-V_1\right)$

(D) the work done $=\pi\left(V_1-V_1\right)^2-\pi\left(P_1-P_1\right)^2$

Q.16

Two containers of equal volume contain the same gas at pressures P_1 and P_2 and absolute temperatures T_1 and T_2, respectively. On joining the vessels, the gas reaches a common pressure P and common temperature T. The ratio P/T is equal to

(A) $\frac{P_1}{T_1}+\frac{P_2}{T_2}$

(B) $\frac{P_1T_1+P_2T_2}{(T_1+T_2)^2}$

(C) $\frac{P_1T_2+P_2T_1}{(T_1+T_2)^2}$

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(D) $\frac{P_1}{2T_1} + \frac{P_2}{2T_2}$

Q.17

Hydrogen gas is filled in a balloon at 20°C. If temperature is made 40°C, pressure remaining same, what fraction of hydrogen will come out?

(A) 0.07

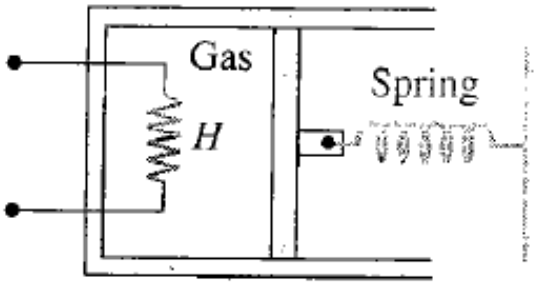
(B) 0.25

(C) 0.5

(D) 0.75

Q.18

An ideal monatomic gas is confined in a cylinder by a spring-loaded piston of cross-section $8 \times 10^{-3}\ \text{m}^2$ Initially the gas is at $300\ \text{K}$ and occupies a volume of $2.4 \times 10^{-3}\ \text{m}^3$ and the spring is in a relaxed state. The gas is heated by a small heater coil H. The force constant of the spring is $8000\ \text{N/m}$ and the atmospheric pressure is $1.0 \times 10^5\ \text{Pa}$. The cylinder and piston are thermally insulated. The piston and the spring are massless and there is no friction between the piston and cylinder. There is no heat loss through heater coil wire leads and thermal capacity of the heater coil is negligible. With all the above assumptions, if the gas is heated by the heater until the piston moves out slowly by $0.1\ \text{m}$, then the final temperature is



(A) $400\ \text{K}$

(B) $800\ \text{K}$

(C) $1200\ \text{K}$

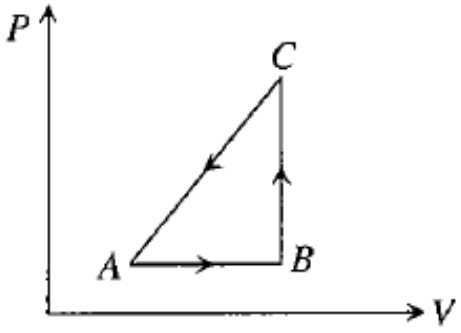
(D) $300\ \text{K}$

Q.19

The $P - V$ diagram of a system undergoing thermodynamic transformation is shown in Fig. 2.96. The work done on the system in going from $A \to B \to C$ is $50\ \text{J}$ and 20cal heat is given to the system. The change

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in internal energy between A and C is



(A) $34\ \text{J}$

(B) $70\ \text{J}$

(C) $84\ \text{J}$

(D) $134\ \text{J}$

Q.20

An air bubble of volume V_0 is released by a fish at a depth h in a lake. The bubble rises to the surface. Assume constant temperature and standard atmospheric pressure P above the lake. The volume of the bubble just before touching the surface will be (density of water is ρ)

(A) V_0

(B) $V_0(\rho gh/P)$

(C) $\frac{V_0}{\left(1+\frac{\rho gh}{P}\right)}$

(D) $V_0\left(1+\frac{\rho gh}{P}\right)$

Q.21

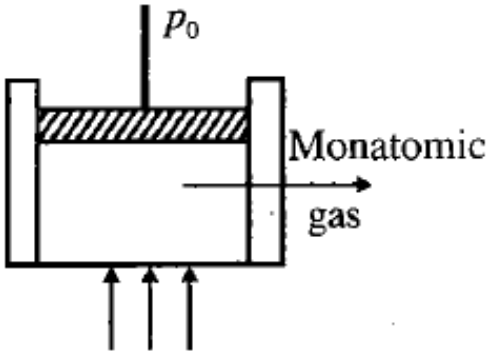
A closed vessel contains $8\ \text{g}$ of oxygen and $7\ \text{g}$ of nitrogen. The total pressure is $10\ \text{atm}$ at a given temperature. If now oxygen is absorbed by introducing a suitable absorbent, the pressure of the remaining gas in atm will be

Q.22

If 50cal of heat is supplied to the system containing 2 mol of an ideal monatomic gas, the rise in temperature

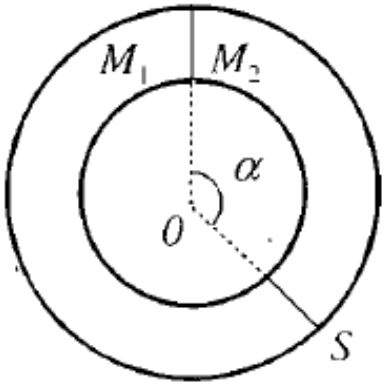
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is $(R = 2\text{cal/mol} - \text{K})$



Q.23

A ring-shaped tube contains two ideal gases with equal masses and molar masses $M_1 = 32$ and $M_2 = 28$. The gases are separated by one fixed partition and another movable stopper S which can move freely without friction inside the ring. The angle α is



Q.24

A stationary cylinder of oxygen used in a hospital has the following characteristics at room temperature $300\ \text{K}$ gauge pressure $1.38 \times 10^7\ \text{Pa}$, volume $16\ \text{L}$. If the flow area, measured at atmospheric pressure, is constant at $2.4\ \text{L/min}$, the cylinder will last for nearly

Q.25

A vessel contains l mole O_2 gas at a temperature T. An identical vessel contains one mole of He gas at a temperature $2T$. Find the ratio of their pressure (He to O_2)

Q.26

If $P - V$ diagram of a diatomic gas is plotted, it is a straight line passing through origin. The molar heat capacity of the gas in the process is nR where n is an integer. Find value of n

Q.27

A vessel contains a mixture of one mole of oxygen and two moles of nitrogen at $300\ \text{K}$. Find the ratio of average rotational kinetic energies per O_2 molecule to per N_2 molecule.

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Q.28

A vessel contains helium, which expands at constant pressure when $15\ \mathrm{kJ}$ of heat is supplied to it. What will be the variation of the internal energy of the gas? (in kJ)

Q.29

A vessel of volume $0.2\ \mathrm{m}^3$ contains hydrogen gas at temeperature $300K$ and pressure 1 bar. Find the heat (in Kcal) required to raise the temperature to $400K$. The molar heat capacity of hydrogen at constant volume is $5\mathrm{cal/mol}K$

Q.30

A vessel of volume $3V$ contains a gas at pressure $4P_0$ and another vessel of volume $2\ \mathrm{V}$ contains the same gas at pressure $1.5P_0$. Both vessels have same temperature. When both vessels are connected by a tube of negligible vol ume, the equilibrium is IP_0, where I is an integer. Find the value of I.

Answers & Solutions

Q.1 Answer:

$\frac{6}{5}$

Solution:

Internal energy of n moles of an ideal gas at temperature T is given by

$$U = \frac{f}{2}nRT \quad (f = \text{ degrees of freedom })$$

$$U_1 = U_2$$

$$\therefore f_1 n_1 T_1 = f_2 n_2 T_2$$

$$\therefore \quad \frac{n_1}{n_2} = \frac{f_2 T_2}{f_1 T_1} = \frac{3\times 2}{5\times 1} = \frac{6}{5}$$

Here, $f_2 =$ degrees of freedom of $\mathrm{He} = 3$ and, $f_1 =$ degrees of freedom of $\mathrm{H_2} = 5$

Q.2 Answer:

$1.77\ \mathrm{kg/m^3}$

Solution:

$$c_p - c_v = \frac{R}{M} = \frac{PV}{TM} = \frac{P}{Td}$$

$$\therefore \quad d = \frac{P}{T\left(c_p - c_v\right)}$$

$$= \frac{1.013\times 10^5}{273(525-315)} = 1.77\ \mathrm{kg/m^3}$$

Q.3 Answer:

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$10^8\pi$J

Solution:

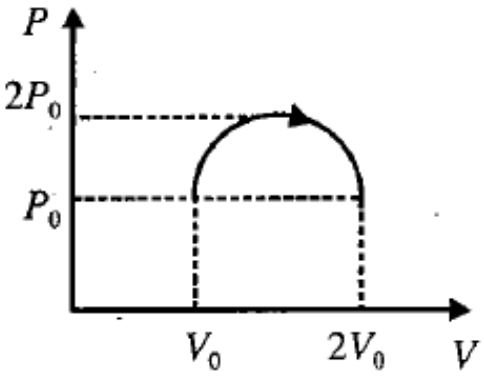
As $\Delta U = 0$ in a cyclic process,

$\Delta Q = \Delta W =$ area of circle $= \pi r^2$

r $= 10$KPa

so $\Delta W = 10^2 \times 10^6 \times \pi$J

Q.4 Answer:



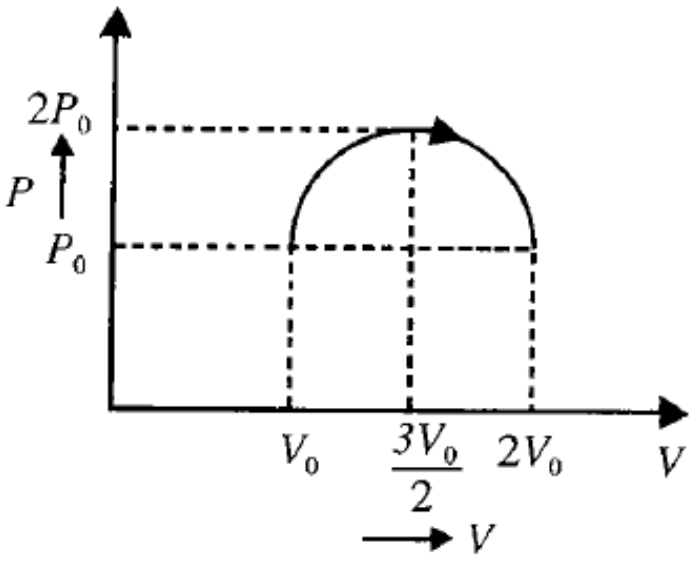
Solution:

The $P - V$ equation is given as

$P = \frac{12P_0}{V_0}V - \frac{4P_0}{V_0^2}V^2 - 7P_0$

putting $P = P_0$, we get

$V^2 - 3V_0V + 2V_0^2 = 0$



$V = \frac{+3V_0 \pm \sqrt{9V_0^2 - 8V_0^2}}{2} = 2V_0$ **or** V_0

At $P = 2P_0$, $V = \frac{3V_0}{2}$

$\frac{dP}{dV} = 0$ at $V = \frac{3V_0}{2}$

$\frac{d^2P}{dV^2} =$ negative

P attains the maximum value at $V = 3V_0/2$

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Q.5 Answer:

$PV^{-2/3} = \text{constant}$

Solution:

Pressure is directly proportional to square of the diameter of the balloon,

$\frac{P_2}{P_1} = \frac{D_2^2}{D_1^2}$ **...(i)**

When $V_2 = 2V_1 \Rightarrow D_2^3 = 2D_1^3 =$

$\Rightarrow \quad \frac{D_2}{D_1} = \left(\frac{V_2}{V_1}\right)^{1/3} \quad \ldots (\text{ii})$

From Eqs. (i) and (ii),

$$\frac{P_2}{P_1} = \left(\frac{V_2}{V_1}\right)^{2/3}$$

$PV^{-2/3} = \text{constant}$

Q.6 Answer:

$\Delta U_5 > \Delta U_3$

Solution:

For paths $1, 2, 3$ and 4, initial temperature of the gas is T_1 and final temperature of the gas is T_2

i.e., $\Delta U_1 = \Delta U_2 = \Delta U_3 = \Delta U_4$

$= nC_V \Delta T = nC_v (T_2 - T_1)\, T_3 > T_2 > T_1$

For path $5, \quad (T_3 - T_1) > (T_2 - T_1)\, \Delta U_5 > \Delta U_3$

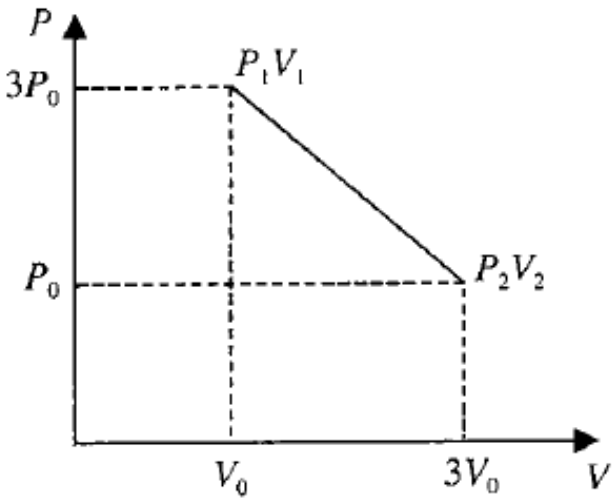
Q.7 Answer:

$\frac{4P_0V_0}{nR}$

Solution:

$$\frac{P - P_1}{P_1 - P_2} = \frac{V - V_1}{V_1 - V_2}$$

$(P - P_1)(V_1 - V_2) = (V - V_1)(P_1 - P_2)$



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$$(P-3P_0)(V_0-3V_0)=(V-V_0)(3P_0-P_0)$$
$$(P-3P_0)(-2V_0)=(V-V_0)(2P_0)$$
$$-2V_0P+6P_0V_0=2VP_0-2P_0V_0$$
$$2VP_0+2V_0P-8P_0V_0=0$$
$$VP_0+\frac{V_0nRT}{V}-4P_0V_0=0$$
$$V^2P_0-4P_0V_0+V_0nRT=0$$
$$T=\frac{P_0\left(-V^2+4VV_0\right)}{V_0nR}$$

For maximum or minimum value of T,

$$\frac{dT}{dV}=-2V+4V_0=0\Rightarrow V=2V_0$$
$$\frac{d^2T}{dV^2}=-2$$

It is negative so

T is maximum at $V=2V_0$

$$T_{\max}=\frac{P_0\left(-4V_0^2+8V_0^2\right)}{V_0nR}=\frac{4P_0V_0}{nR}$$

Q.8 Answer:

-10 kJ

Solution:

$$6T_1=3T_2=2T_4=T_3=1800\text{ K}$$
$$T_1=300\text{ K};\quad T_2=600\text{ K}$$
$$T_4=900\text{ K};\quad T_3=1800\text{ K}$$

$4\to1$ and $2\to3$ are isochoric processes in which work done $=0$

$$W_{12}=P(V_2-V_1)=nR(T_2-T_1)$$
$$=2\times R(600-300)=600R$$
$$W_{34}=P(V_4-V_3)=nR(T_4-T_3)$$
$$=2\times R(900-1800)=-1800R$$
$$W_{\text{Total}}=600R-1800R=-1200R=-10000\text{ J}$$

Q.9 Answer:

$2\sqrt{2}:1:2$

Solution:

b. Let the initial pressure of the three samples be P_A, P_B and P_c, then

$$P_A(V)^{3/2}=(2V)^{3/2}P\quad(\because P_B=P)$$
$$\text{or } P_A=P(2)^{3/2}$$
$$P_C(V)=P(2V)$$
$$\text{or } P_C=2P$$
$$\therefore P_A:P:P_C$$
$$=(2)^{3/2}:1:2=2\sqrt{2}:1:2$$

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Q.10 Answer:

$m = 7n$

Solution:

$C_p - C_v = \mathrm{m}$, **for hydrogen** $(M_1 = 2)$ $C_p - C_v = n$, **for nitrogen** $(M_2 = 14)$

For hydrogen, $C_p - C_v = \frac{1}{M_1}\frac{dQ}{dT} = m$

For nitrogen, $C_p - C_v = \frac{1}{M_2}\frac{dQ}{dT} = n$

$\therefore \frac{m}{n} = \frac{\frac{1}{M_1}\frac{dQ}{dT}}{\frac{1}{M_2}\frac{dQ}{dT}} = \frac{M_2}{M_1} = \frac{14}{2} = 7$

$\therefore \quad m = 7n$

Q.11 Answer:

$\frac{4}{9}\%$

Solution:

For an adiabatic process, $PV' = K$

Here, $\gamma = 3/2$ and $K =$ constant

$\therefore PV^{3/2} = K$

$\log P + \frac{3}{2}\log V = \log K$

$\frac{\Delta P}{P} + \frac{3}{2}\frac{\Delta V}{V} = 0$

$\therefore \frac{\Delta V}{V} = -\frac{2}{3}\frac{\Delta P}{P}$

$\frac{\Delta V}{V} \times 100 = -\left(\frac{2}{3}\right)\left(\frac{\Delta P}{P} \times 100\right) = -\frac{2}{3} \times \frac{2}{3} = -\frac{4}{9}$

Therefore volume decreases by about (4/9)%

Q.12 Answer:

300 K

Solution:

It is free expansion, temperature will remain constant.

Q.13 Answer:

$(1/\eta) - 1$

Solution:

$\eta = 1 - \frac{T_2}{T_1}$

$\omega = \frac{T_2}{T_1 - T_2} = \frac{T_2/T_1}{1-(T_2/T_1)} = \frac{(1-\eta)}{\eta} = \frac{1}{\eta} - 1$

Q.14 Answer:

80 J

Solution:

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Work done by the gas al constant pressure

$$\Delta W = P\Delta V$$
$$= \left(1 \times 10^5 \mathrm{Nm}^{-2}\right)(1000 - 800) \times 10^{-6}\ \mathrm{m}^3 = 20\ \mathrm{J}$$
$$\Delta U = \Delta Q - \Delta W$$
$$\Delta U = 100\ \mathrm{J} - 20\ \mathrm{J} = 80\ \mathrm{J}$$

Q.15 Answer:

the work done $= -(\pi/4)(P_2 - P_1)(V_2 - V_1)$

Solution:

The work done = area of $P - V$ diagram

$a = \frac{V_2 - V_1}{2}, \quad b = \frac{P_2 - P_1}{2}$

$W = -\pi\left(\frac{V_2 - V_1}{2}\right)\left(\frac{P_2 - P_1}{2}\right)$

But the cyclic process is anticlockwise. Hence, the work done is negative.

Q.16 Answer:

$\frac{P_1}{2T_1} + \frac{P_2}{2T_2}$

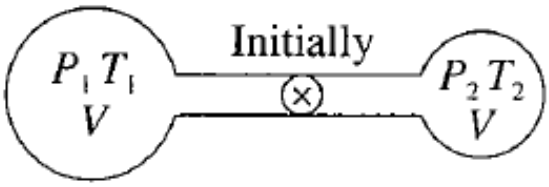
Solution:

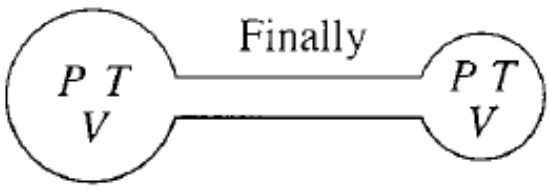
Number of moles in the first vessel

$\mu_1 = \frac{P_1 V}{RT_1}$

Number of moles in the second vessel

$\mu_2 = \frac{P_2 V}{RT_2}$





If both vessels are joined together, then quantity of gas remains same, i.e., $\mu = \mu_1 + \mu_2$

$$\frac{P(2V)}{RT} = \frac{P_1 V}{RT_1} + \frac{P_2 V}{RT_2}$$
$$\frac{P}{T} = \frac{P_1}{2T_1} + \frac{P_2}{2T_2}$$

Q.17 Answer:

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0.07

Solution:

As $V \propto T$

$\therefore \quad \frac{V_2}{V_1} = \frac{T_2}{T_1}$

$\Rightarrow V_2 = \left(\frac{313}{293}\right) V_1$

Fraction of gas that comes out

$= \frac{V_2 - V_1}{V_1} = \frac{\left(\frac{313}{293}\right)V_1 - V_1}{V_1} = \frac{20}{293} = 0.07$

Q.18 Answer:

800 K

Solution:

$V_1 = 2.4 \times 10^{-3}\ \mathrm{m}^3, \quad P_1 = P_0 = 10^5\ \mathrm{N/m}^2$

and $\quad T_1 = 300\ \mathrm{K} \quad$ (given)

If area of cross section of piston is A and it moves through distance x, then increment in volume of the gas $= Ax$.

If force constant of a spring is k, then force $F = kx$ and pressure $= F/A = kx/A$

$V_2 = V_1 + Ax = 2.4 \times 10^{-3} + 8 \times 10^{-3} \times 0.1 = 3.2 \times 10^{-3}$

and $P_2 = P_0 + \frac{kx}{A} = 10^5 + \frac{8000 \times 0.1}{8 \times 10^{-3}} = 2 \times 10^5$

From ideal gas equation

$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$

$\Rightarrow \frac{10^5 \times 2.4 \times 10^{-3}}{300} = \frac{2 \times 10^5 \times 3.2 \times 10^{-3}}{T_2}$

$\Rightarrow T_2 = 800\ \mathrm{K}$

Q.19 Answer:

134 J

Solution:

Heat given, ΔQ

$= 20\mathrm{cal} = 20 \times 4.2 = 84\ \mathrm{J}$

Work done,

$\Delta W = -50\ \mathrm{J}$

(as the process is anticlockwise)

By the first law of thermodynamics,

$\Delta U = \Delta Q - \Delta W = 84 - (-50) = 134\ \mathrm{J}$

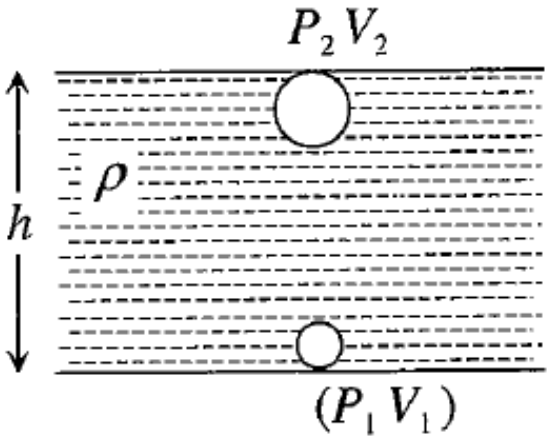
Q.20 Answer:

$V_0\left(1 + \frac{\rho g h}{P}\right)$

Solution:

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According to Boyle's law, multiplication of pressure and volume will remain constant at the bottom and top.



If P is the atmospheric pressure at the top of the lake and the volume of bubble just before touching surface is V then from $P_1V_1 = P_2V_2$

$$(P + h\rho g)V_0 = PV \Rightarrow V = \left(\frac{P + h\rho g}{P}\right)V_0$$

$$\therefore V = V_0\left[1 + \frac{\rho gh}{P}\right]$$

Q.21 Answer:

—

Solution:

From Dalton's law, final pressure of the mixture of nitrogen and oxygen is

$$P = P_1 + P_2$$
$$= \frac{\mu_1 RT}{V} + \frac{\mu_2 RT}{V} = \frac{m_1}{M_1}\frac{RT}{V} + \frac{m_2}{M_2}\frac{RT}{V}$$
$$= \frac{8}{32}\frac{RT}{V} + \frac{7}{28}\frac{RT}{V} = \frac{RT}{2V} \Rightarrow 10 = \frac{RT}{2V}$$

When oxygen is absorbed then for nitrogen let pressure is

$$P = \frac{7}{28}\frac{RT}{V}$$
$$\Rightarrow P = \frac{RT}{4V}$$

From Eqs. (i) and (ii), we get pressure of the nitrogen $P = 5$ atm

Q.22 Answer:

—

Solution:

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$$\frac{C_v}{C_p} \times Q = nC_v dT$$

$$dT = \frac{Q}{nC_p} = \frac{50}{2 \times \frac{5}{2} \times R} = 5 \text{ K}$$

Q.23 Answer:

—

Solution:

$PV = \frac{m}{M}RT$ **(for ideal gas)**

$\therefore MV = \frac{mRT}{P}$

In the position of equilibrium of stopper S**,**

$$P_1 = P_2, \quad T_1 = T_2, \quad m_1 = m_2$$

$$\therefore \quad MV = \text{ constant}$$

$$M_1V_1 = M_1V_2$$

$$\Rightarrow A \times 32(360 - \alpha) = 28a \times A$$

$$\alpha = 192^\circ$$

Q.24 Answer:

—

Solution:

$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

$\frac{\left(1.38 \times 10^7 \text{ Pa}\right)(16 \text{ L})}{300 \text{ K}} = \frac{\left(10^5 \text{ Pa}\right)\left(2.4 \frac{\text{L}}{\text{m}} t\right)}{300 \text{ K}}$

$t = \frac{1.38 \times 10^7 \times 16}{10^5 \times 2.4} = 920 \text{ min}$

$= \frac{920}{60} \text{ h} = 15 \text{ h}$ **(approx.)**

Q.25 Answer:

—

Solution:

Using $P = \frac{nRT}{V}$**, we get**

$\frac{P_{\text{He}}}{P_{\text{O}}} = \frac{2T}{V}\frac{V}{T} = 2$

Q.26 Answer:

—

Solution:

$P \propto V$ **since graph is a straight line** $PV^{-1} = \text{constant}$

For the process $PV^x =$ **Constant, molar heat capacity is given by** $C = \frac{R}{(\gamma - 1)} + \frac{R}{(1 - x)}$

B**ut here** $\gamma = 1.4$ **and** $x = -1$

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$\text{SO}, C = \frac{R}{(1.4-1)} + \frac{R}{(1+1)} = 3R = nR$ **(Given)**

So, $n = 3$

Q.27 Answer:

—

Solution:

Both gases will have same number of rotational degree of freedom (equal to 2).

Since both are diatomic and are at the same temperature.

Average rotational kinetic energy per molecule for both the gases $= 2\left(\frac{1}{2}KT\right)$.

Required ratio $= 1$

Q.28 Answer:

—

Solution:

Here $\Delta Q = 15000\ \text{J}$ **(given)**

In an isobaric process

$\Delta Q = nC_p\Delta T, \quad \Delta U = nC_v\Delta T\ (\text{always})$

$\frac{\Delta U}{\Delta Q} = \frac{nC_V\Delta T}{nC_p\Delta T} = \frac{1}{\gamma} \Rightarrow \Delta U = \frac{\Delta Q}{\gamma}$

or $\Delta U = \frac{3}{5} \times 15 = 9\ \text{kJ}$

Q.29 Answer:

—

Solution:

As here volume of gas remains constant, $(\Delta Q)_V = \mu C_v \Delta T,$

Here $C_v = 5\text{cal/molK}$

And $\Delta T = (400 - 300) = 100\ \text{K}$

And so for ideal gas $PV = \mu RT,$

$\mu = \frac{(10)^5 \times (0.2)}{8.31 \times 300} = 8\ \text{mol}$

$(\Delta Q)_V = 8 \times 5 \times 100 = 4\text{kcal}$

Q.30 Answer:

—

Solution:

No change of temperature

So , $(n_1 + n_2)_i = (n_1 + n_2)_f$

$\frac{P_1V_1}{RT} + \frac{P_2V_2}{RT} = \frac{PV_1}{RT} + \frac{PV_2}{RT}$

$\text{SO}, P = \frac{P_1V_1 + P_2V_2}{V_1 + V_2} = \frac{(4P_0)(3V) + (1.5P_0)(2V)}{5V} = 3P_0 = IP_0$ **(given)**

So, $I = 3$

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